

TOKYO, Japan

WMO: 476620

Lat: 35.6917N

Lon: 139.7511E

Elev: 79

StdP: 14.65

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 33.2	(c) 34.7	(d) 5.1	(e) 7.2	(f) 41.7	(g) 7.9	(h) 8.3	(i) 41.6	(j) 19.8	(k) 44.2	(l) 18.0	(m) 44.9	(n) 7.0	(o) 340	(p) 0.409

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 10.9	(c) 92.6	(d) 78.0	(e) 90.6	(f) 77.3	(g) 88.7	(h) 76.5	(i) 79.9	(j) 88.3	(k) 78.9	(l) 86.9	(m) 78.0	(n) 85.7	(o) 9.0	(p) 150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.8	144.9	84.2	76.6	139.0	83.6	75.6	134.3	83.4	43.6	88.3	42.5	87.1	41.5	85.9	84.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 18.4	(b) 16.0	(c) 14.0	(e) 30.3	(f) 97.0	(g) 1.7	(h) 2.6	(i) 29.0	(j) 98.9	(k) 28.0	(l) 100.4	(m) 27.1	(n) 101.8	(o) 25.8	(p) 103.7		
			DB													
			WB													
				30.3	97.0	1.7	2.6	29.0	98.9	28.0	100.4	27.1	101.8	25.8	103.7	
				24.3	81.3	1.5	1.5	23.2	82.4	22.3	83.3	21.5	84.2	20.4	85.3	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	62.2	43.1	44.2	49.8	59.0	67.2	72.7	80.2	82.4	75.9	66.4	56.4	47.6
	DBStd	14.35	3.69	4.63	5.72	5.91	4.80	4.70	5.24	4.56	5.68	5.04	4.90	4.40
	HDD50	579	216	173	77	6	0	0	0	0	0	0	5	102
	HDD65	2806	679	583	472	193	32	3	0	0	2	42	259	541
	CDD50	5023	3	10	70	275	534	681	937	1005	777	507	198	26
	CDD65	1775	0	0	0	12	101	234	472	540	329	85	2	0
	CDH74	15260	0	0	1	31	345	1209	4781	6083	2541	268	1	0
	CDH80	5315	0	0	0	1	39	237	1788	2505	715	30	0	0
(14) Wind	WSAvg	7.0	7.0	7.4	7.7	7.8	7.4	6.6	6.8	6.8	6.9	6.6	6.2	6.4
(15) Precipitation	PrecAvg	60.9	2.4	2.5	4.4	4.9	5.4	6.3	5.9	5.7	8.6	8.2	4.3	2.4
	PrecMax	80.3	5.6	5.6	7.8	9.2	10.1	12.6	12.8	14.0	18.2	26.9	10.1	7.8
	PrecMin	45.0	0.1	0.2	1.4	1.7	2.4	1.7	1.2	0.6	1.8	0.7	0.3	0.0
	PrecStd	8.6	1.6	1.4	1.5	2.0	2.0	2.3	3.6	3.7	4.1	5.8	2.4	1.6
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	60.0	65.4	70.4	77.5	84.4	88.7	94.7	95.9	91.5	83.6	72.6	65.7
		MCWB	50.8	54.3	55.9	61.4	66.4	73.7	77.8	78.3	75.7	70.4	60.6	56.0
	2%	DB	55.7	59.4	66.4	73.8	80.1	85.3	92.2	93.3	88.7	79.5	69.5	60.7
		MCWB	45.6	49.2	54.5	60.0	65.4	71.7	77.7	78.5	75.3	68.4	59.1	49.7
	5%	DB	52.6	55.2	63.1	71.1	77.6	82.6	90.3	91.4	86.4	76.9	67.0	57.7
		MCWB	42.3	45.7	52.9	59.4	64.3	70.8	77.2	78.0	74.8	66.9	57.3	48.1
	10%	DB	50.3	52.3	60.1	68.6	75.4	80.3	88.1	89.4	84.0	74.4	64.6	55.2
		MCWB	40.3	42.5	50.6	57.5	63.3	69.9	76.5	77.3	73.7	64.8	55.7	46.2
	0.4%	WB	53.1	57.6	60.5	68.2	71.8	76.4	81.1	82.0	79.2	73.8	64.7	58.1
		MCDB	58.7	64.3	65.3	71.0	77.9	84.8	90.2	91.2	86.3	79.4	69.5	63.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	47.1	50.7	57.6	63.9	69.0	74.7	79.4	80.4	77.6	71.4	61.8	52.9
		MCDB	53.6	57.5	64.0	70.7	75.4	82.3	88.1	88.8	84.3	77.4	66.5	58.7
	5%	WB	44.0	46.7	54.8	61.3	67.4	73.2	78.4	79.4	76.7	69.1	59.6	49.6
		MCDB	50.9	53.3	61.7	68.5	74.1	79.9	86.7	87.4	83.4	74.7	64.6	55.6
	10%	WB	41.7	44.1	51.5	59.5	65.7	71.7	77.5	78.5	75.7	66.6	57.5	47.3
		MCDB	48.8	50.8	58.5	66.1	72.6	77.9	85.4	86.3	82.5	72.4	62.8	53.6
(35) Mean Daily Temperature Range	5% DB	MDBR	12.3	12.6	13.3	13.5	12.5	10.7	10.7	10.9	10.3	10.5	11.4	11.9
		MCDBR	16.0	17.1	17.2	16.3	15.9	13.7	13.2	13.0	12.2	13.4	14.5	15.4
		MCWBR	10.9	12.7	12.0	9.4	7.9	5.8	5.1	4.7	5.2	7.9	10.1	10.7
	5% WB	MCDBR	15.0	16.0	15.8	14.2	13.5	11.4	12.3	12.2	11.1	12.1	12.8	14.1
		MCWBR	11.6	13.2	12.9	10.1	8.5	5.8	5.1	5.0	5.4	8.3	10.5	11.3
(40) Clear-Sky Solar Irradiance	taub		0.380	0.429	0.497	0.548	0.572	0.608	0.600	0.588	0.520	0.488	0.428	0.383
	taud		2.238	2.096	1.937	1.858	1.857	1.837	1.905	1.922	2.074	2.090	2.171	2.252
	Ebn at Noon		252	251	244	239	235	226	227	226	235	230	232	242
	Edh at Noon		34	44	56	64	65	66	62	59	49	44	36	32
(44) All-Sky Solar Radiation	RadAvg		879	1041	1299	1514	1626	1435	1555	1563	1214	962	797	767
	RadStd		61	102	82	148	171	133	261	149	108	101	75	59

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (38 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page